

ANAND P

QUANTITATIVE RESEARCHER & ALGO DEVELOPER

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Research portfolio: [quant-research-portfolio.vercel.app](#) | GitHub: [github.com/Anand3643](#)

Open to quantitative trader & systematic researcher seats · 100% remote · IST (UTC+5:30), adaptable to US / London / Asia sessions

PROFESSIONAL SUMMARY

Quantitative researcher and algorithmic trading developer who builds the whole chain — data, backtest, parameter sweep, then the production software that runs the survivors unattended. Research spans Nifty 500 equities, Nasdaq 100, Indian F&O, XAU/USD and a twelve-year Bitcoin study, every figure regenerated from a file on disk rather than typed in. Methodology is the emphasis: zero-lookahead signal lagging, costs and cash yield modelled explicitly, medians reported across full parameter grids rather than best rows, and losing configurations published alongside winners. Python, MQL5, Pine Script, FastAPI, PostgreSQL, MetaTrader 5.

SELECTED QUANTITATIVE RESEARCH

Bitcoin (BTC-USD) Multi-Timeframe Strategy Study — 120-strategy universe Sep 2014 – Sep 2026 · 4,370 bars
120 daily systems across six families plus **77** weekly-over-daily confluence pairs, a 1D/3D/1W resolution benchmark and a six-regime stress test. Zero-lookahead lagging (T-1 daily, W-1 weekly) with idle cash compounded at 4.0%. Best system: **2.00** Sharpe, **83.2%** CAGR — but the defensible result is drawdown control at **-41.5%** against buy-and-hold's **-83.4%**. Ranking documented as in-sample.

Momentum Parameter Sweep — Nifty 500 8,640 configurations
8,640 portfolio configurations swept across eight axes — scoring, SMA filter, trailing stop, portfolio size, weighting, rebalance interval, exit buffer, hedging. Reported the **median Sharpe of 1.17** across the full grid rather than the best row at 2.00, because the best row is selection, not evidence.

15-Year Breakout Similarity Study — Nifty 500 2010 – 2026 · 4,101 sessions
32,722 breakout events classified over a 60-day forward horizon: **9,966** reached +20%, 1,932 reached +50%, 468 doubled, and **11,288** failed below a 5% peak — a 34.5% failure rate reported as prominently as the win rate. Isolated the entry-state features that separate the two: distance from the 52-week high, extension above the 200-day SMA, prior 3-month momentum.

Signal-Engine Comparison — controlled design 192,482 trades
10 entry engines specified, implemented and measured over **484** symbols and **192,482** trades on one identical universe with identical exits, so the entry signal was the only variable. Ranked on per-symbol medians, not means, so a few runaway winners cannot describe a strategy.

Nasdaq 100 Momentum Rotation — Setup F, top 3 bi-weekly 2015 – 2024 · 2,516 sessions
41.72% CAGR, **-39.09%** max drawdown, Calmar **1.07** against **18.30%** for QQQ over the same window — \$100,000 into \$3.26M across 284 trades at 5 bps per side. Published as the sixth of six setups tested, with the survivorship-biased universe stated plainly as the largest qualifier on the result.

XAU/USD Expert Advisors — automated MT5 test harness 42 runs · 83,917 trades
PowerShell pipeline that compiles every advisor, generates a tester config per strategy and timeframe, runs MetaTrader 5 headless, parses the HTML reports and ranks them — unattended, start to finish. **14** strategies, **3** timeframes, forced-identical lot size; **20 of 42** runs profitable, all 42 published.

F&O Opening-Range Study — exit-method comparison 7 methods · common trade set
7 exit methods evaluated on one common trade set; **4 lost money**, including a **63.2%** win-rate BTST method with negative expectancy (profit factor 0.98) — a direct demonstration that hit rate alone is not evidence of edge. The losing methods are published.

PRODUCTION SYSTEMS & ENGINEERING

FastAPI trading & reporting service — **15,264** lines across **33** services and **39** endpoints, running unattended and publishing **75** automated end-of-day reports to Telegram on a 16:10 IST loop.

Backtesting infrastructure — vectorised pandas/NumPy engines with explicit cost models, position sizing and regime filters; generated data pipelines so every figure on the research site is recomputed from source CSVs rather than hand-entered, and site and reports cannot drift apart.

Strategy library & publication — **46** MQL5 expert advisors and **7** Pine Script indicators, a source-code audit of 44 advisors flagging 8 high-risk martingale/grid designs, and **24** research reports (12 published in full).

TECHNICAL SKILLS

Programming & Data	Python, pandas, NumPy, asyncio, SQL, vectorised backtesting
Trading & Quant	Systematic strategy development, parameter sweeps, walk-forward and regime stress testing, risk & performance measurement (Sharpe, Calmar, drawdown, exposure), portfolio construction, equity research
Trading Platforms	MQL5, MetaTrader 5 (Strategy Tester automation), Pine Script
Backend & Data	FastAPI, PostgreSQL, REST APIs, scheduled unattended services
Web	React.js, JavaScript, HTML, CSS, Node.js, Redux, MongoDB

PROFESSIONAL EXPERIENCE

Ramraj Enterprises | Equity Trader Aug 2025 – Present · Chennai
• Trade Indian equities while running the independent quantitative research programme above — strategy design, backtesting and the production systems that execute and report on them.

Human Resources roles | The Healer Foundation · Takecare Recruitment Consultancy · ILP Overseas Jan 2023 – Jul 2025

Teachnook | Web Development Intern Jan 2023 – Feb 2023
• Built React applications with Redux and Redux Forms, and object-oriented Node.js services, working alongside backend developers on live client sites.

EDUCATION & CERTIFICATIONS

Bharath University — MBA, Human Resources & Finance (Jul 2022) · **Surya Group of Institutions** — BE, Civil Engineering, 81% (2019 – Apr 2022) · **Surya Polytechnic College** — Diploma, Civil Engineering, 83% (2016 – Apr 2019)

Certifications — **GUVI** 2023 (Machine Learning 101, MySQL, MongoDB, AWS, JavaScript, Modern HTML & CSS) · **Udemy** Complete 2023 Web Development Bootcamp, 66 hrs · **Teachnook** Web Development course · DMO, Diploma Robotics, AutoCAD — IDs verifiable on the portfolio.